Bachelor Thesis

Nsak as Framework for Scenario Based Network Security

Course of study	Bachelor of Science in Computer Science
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Abstract

One-paragraph summary of the entire study – typically no more than 250 words in length (and in many cases it is well shorter than that), the Abstract provides an overview of the study.

Contents

1. Introduction

Computer networks are constantly exposed to evolving cyber threats ...

The *Network Swiss Army Knife (NSAK)* is a framework designed to execute orchestrated security scenarios consisting of reusable attack drills in a controlled network environment. The framework aims to support automated security testing and adversary emulation on network edge devices.

The development of the NSAK framework originates from previous project work, in which the core architecture and the basic scenario execution model were developed. This thesis builds upon this foundation and extends the framework by investigating how AI agents can support automated red and blue team scenarios.

...

1.1. Motivation

Hypothesis: AI can enhance red and blue team simulations in a network environment by generating and executing scenarios within the NSAK framework and assisting in the evaluation of detection results.

1.2. Project Goals

For a goal to be considered **S.M.A.R.T.**, it must fulfill the following criteria:

- **Specific**: What do we want to achieve?
- **Measurable**: What are the success criteria?
- **Achievable**: How do we plan to achieve this goal?
- **Relevant**: Why is it important to achieve this goal?
- **Time-bound**: Goals categorized as *must* have to be achieved within the thesis, while *should* or *could* goals are considered optional or nice to have.

1.2.1. Research Goals

Compare Frameworks (must)

To identify the USP (unique selling point) of the NSAK framework, we will analyze and compare existing security emulation frameworks. First, relevant frameworks such as MITRE Caldera and Atomic Red Team will be identified and reviewed. Based on this analysis, the identified concepts will be compared with the architecture of the NSAK framework. The comparison will focus on how scenarios and drills are defined, how configuration and parameter handling are structured, how reusable components are organized, how automation is achieved, and which AI capabilities they provide.

The analysis and comparison will be based on the following properties:

- Concepts
- Modularity
- Configurability
- Automation
- AI Capabilities

Success criteria:

- At least two security emulation frameworks are identified and reviewed
- A matrix comparing the specified properties of the selected frameworks is created
- The USP of the NSAK framework is identified

AI Integration (must)

The goal of this research is to investigate how AI (artificial intelligence) can support automated security testing and adversary emulation. The research evaluates potential integration points where AI could enhance the scenario execution, automation, decision-making, and analysis capabilities of the NSAK framework.

Success criteria:

- How is AI currently used in cybersecurity automation?
- Which tasks in adversary emulation can be supported or automated by AI?
- How could AI interact with the NSAK scenarios and drills (e.g., Agents, MCP)?

- What architectural changes would be required to integrate AI into NSAK?

1.2.2. Framework

Configuration Management (must)

The goal is to extend the NSAK framework, so that scenarios and drills can expose configurable parameters to increase the flexibility of scenarios and reusability of drills. Parameters can be provided via command-line arguments or YAML files and are resolved by the framework into a unified run configuration used for scenario execution.

Success criteria:

- Scenarios and drills can expose configurable parameters
- The available parameters can be listed by the CLI with the `--help` option
- Parameters can be provided via CLI arguments or YAML files
- The framework resolves the inputs into a run configuration (internal data structure).

REST API (TBD)

Adding a REST API to the NSAK framework was already proposed in the predecessor project, as it greatly enhances interoperability with other tools. The priority of this goal depends heavily on the outcome of the goal 1.2.1 and on whether we want to fulfill the goal 1.2.2, which is prioritized as “could”. If the evaluated AI integration requires a REST API (e.g., MCP), this goal needs to be prioritized as “must”; it will be prioritized as “could”.

Success criteria:

- A REST API specification with feature parity to the CLI is created.
- The REST API is implemented according to the specification.
- The API is documented (Open API) and usable by external clients.
- The REST API remains optional and does not replace the CLI-based interaction.
- Authentication and Authorization are explicitly out of scope.

MCP Server (TBD)

The MCP Protocol allows the AI to interact with applications in a standardized way - and could therefore be necessary for implementing AI-based scenarios. If the outcome of the research task 1.2.1 highlights the necessity to specify/expose an MCP-Server to interact seamlessly with the NSAK framework, the implementation is a (must), otherwise a (could).

- A MCP-Server is specified and implemented
- Documentation how to connect to the MCP is provided.
- A form of authentication is provided.

Web GUI (could)

The goal is to design and implement a web-based graphical user interface that allows users to interact with the NSAK framework in a user-friendly way. The Web GUI will enable management of scenarios and drills, triggering scenario execution, and visualization of execution results and system status.

Success criteria:

- A simple Web GUI that interacts with the REST API has been implemented.
- The Web GUI displays a list of available scenarios and their states.
- The scenario's results can be viewed on the Web GUI
- Scenarios can be configured and executed via the Web GUI
- Authentication and authorization are explicitly out of scope.

1.2.3. Scenarios

AI – Purple Team (must)

The goal is to implement artificial intelligence according to the result of 1.2.1 and to explore the use of AI to support red, blue, and purple team exercises in the NSAK framework. The AI will be evaluated for its potential to assist in executing attacks in a scenario and analyzing detection results. The findings will help assess how AI could enhance and coordinate red and blue team activities across a network via the NSAK device.

Success criteria:

- The AI is integrated into NSAK as a scenario or as part of the framework, depending on the result of 1.2.1.
- The AI can execute CLI or Python tools to directly or indirectly interact with the attached network.
- An operator can interact with or stop the running AI.
- The result artifacts are stored and can be presented or reused for further scenarios.

Reproducible Test Environment (must)

The goal is to design and implement a reproducible and safe test environment for the NSAK framework that enables consistent execution and evaluation of scenarios and drills. The environment should provide a stable basis for testing and comparison of manual and AI-assisted scenarios.

Success criteria:

- The test environment can be set up reproducibly using a defined configuration or setup procedure with Infrastructure as code (e.g., Ansible)
- NSAK scenarios and drills can be executed consistently within the environment.
- The environment supports both manual and AI-assisted scenarios.
- Scenario results can be collected and analyzed for evaluation purposes.

Network Discovery – Red Team (must)

The goal is to implement a reference red team scenario for network discovery within the NSAK framework. The scenario should simulate reconnaissance activities to identify hosts, services, and network characteristics, and the results should be comparable to the AI-Agent approach.

Success criteria:

- A red team network discovery scenario is implemented in the NSAK framework.
- Discovered network information is collected as artifacts.
- The scenario can be executed reproducibly within the test environment.
- The results can be compared with those generated by the AI-assisted approach.

Intrusion Detection – Blue Team (should)

The goal is to define and implement a blue team reference scenario for intrusion detection within the NSAK framework. The scenario should allow observation and evaluation of suspicious network activity to assess detection capabilities in the test environment and compare the results with those of the AI approach.

Success criteria:

- A blue-team intrusion-detection scenario is implemented in the NSAK framework.
- Detection events or alerts can be observed and documented during scenario execution.
- Detection results can be compared with results from AI-assisted scenario execution.

1.2.4. Evaluation Goals

AI Scenario (must)

The objective is to determine the effectiveness of the AI-based scenario introduced by the goal 1.2.3 within the network environment defined by the goal 1.2.3 The evaluation will assess whether the AI scenario successfully executed the expected red and blue team activities for the provided prompts and the quality of the results.

Success criteria:

- A methodical approach for the assessment is evaluated and documented.
- The results of the AI scenario for red and blue team activity are assessed.
- There is a conclusion on the effectiveness of the AI scenario for red and blue team activities.

AI vs Engineered Scenarios (must/should)

The goal of this evaluation is to compare the quality of AI-based red (network reconnaissance) (must) and blue (network intrusion detection) (should) team scenarios with that of manually engineered scenarios.

The comparison focuses on criteria such as execution reliability, reproducibility, and coverage. The results will help determine the potential benefits and limitations of AI-supported scenarios and provide insights into how AI could enhance future versions of the NSAK framework.

Success criteria:

- A methodical approach for the comparison is evaluated and documented.
- The results of the AI scenario are compared against the respective results of the red team scenario (must)
- The results of the AI scenario are compared against the respective results of the blue team scenario (should)
- There is a conclusion on the effectiveness of the AI scenario compared to manually engineered scenarios.

1.2.5. Out of Scope / Optional Feature

At the project management meeting on 05.03.26 (see Table **??** in Appendix A.7), the initial project scope was discussed. During planning, the team proposed using AI-based scenarios and comparing them with the implemented versions, which required reassessing certain milestones.

Running AI-based scenarios does not require a REST API or GUI. A reproducible test setup and further research are needed to properly evaluate this approach.

2. Related Research

2.1. Comparison Nsak - Caldera

3. Concepts - Framework

4. Methodic

5. Implementation

6. Evaluation

7. Discussion

8. Conclusion

Bibliography

List of Figures

List of Tables

Glossary

This document is incomplete. The external file associated with the glossary 'main' (which should be called `thesis.gls`) hasn't been created.

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  ```

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A. Project Management

A.1. Agile Management Process

The project followed an interactive development approach using Gitlab for issue tracking and planning. Work items were organized into epics, milestones and issues. Each iteration lasted two weeks.

Issue Lifecycle

The workflow used for ticket tracking consisted of the following stages

- Backlog
- Todo
- Q&A
- Done

A.2. Definition of Ready (DoR)

An ticket is considered ready for implementation when the following criteria are fulfilled.

Clarity

- Clear and precise title
- Objective or problem described in two to three sentences
- Scope clearly defined

Scope

- Affected module specified (drill, scenario, core, container, frontend)
- External dependencies identified

Validation

- Testable acceptance criteria defined
- Expected behavior clearly specified

A.3. Definition of Done (DoD)

A ticket is considered done when all the following criteria are fulfilled.

Implementation

- All acceptance criteria fulfilled
- Container execution verified
- Pre-commit hooks pass
- Logging and error handling consistent

Testing

- Feature or bug fix tested
- Acceptance criteria verifiable (automated or manual)
- Relevant edge cases considered
- Execution reproducible

Documentation

- README updated if required
- Architecture changes documented
- Thesis relevance briefly noted
- Limitations documented
- Required documentation stored for thesis or appendix

A.4. Project Structure

Milestones Milestones represent crucial steps toward the overall project goal.

Epics Epics consist of multiple stories and contribute to fulfilling milestones.

Stories Each story is assigned a priority, sprint, milestone and epic. Stories must meet the Definition of Ready before implementation.

A.5. Planning and Estimation

The following estimation strategy was used:

- 1 hour corresponds to 1 weight point
- Iteration length: 2 weeks
- Sprint Planning duration: 1.5 hours per iteration
- Sprint Review duration: 1 h per iteration
- Sprint Retro duration: 1 h per iteration

maybe sub-section would be better

Roadmap The introduced project structure is illustrated in the roadmap shown in Figure A.1. The structure shown in Label A.4 helps to visualize the project goals and keeps track of the progress towards achieving goals and milestones. Furthermore, the visualization helps stakeholders to grasp the current state of the project.

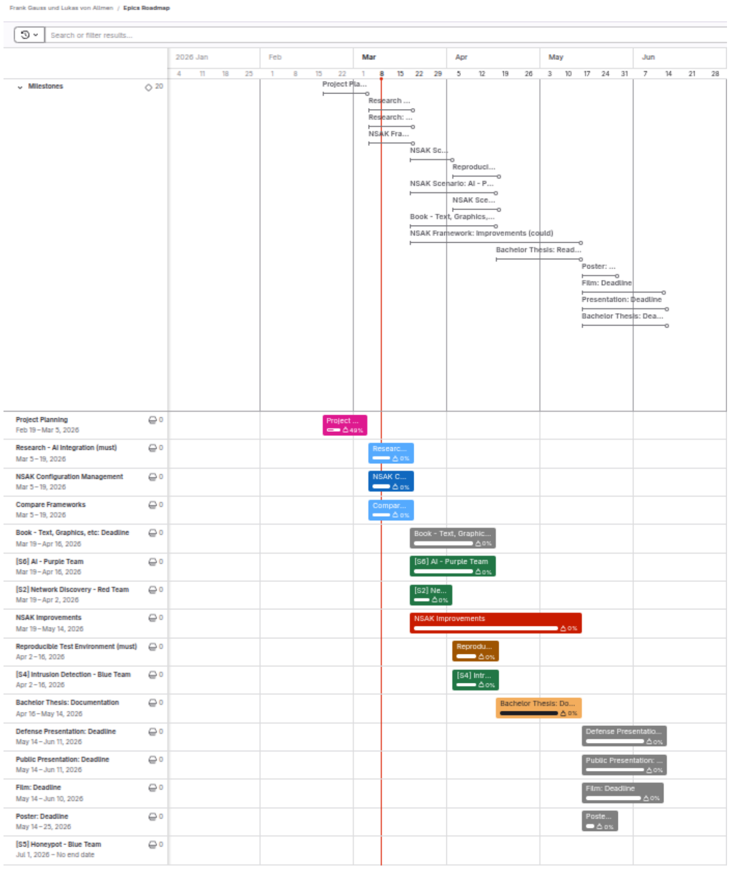

Figure A.1.: Roadmap

A.5.1. Risk Assessment

Risk Identification

Risk Analysis

Risk Evaluation / Treatment

Risk treatment strategies:

- Mitigate
- Transfer
- Accept
- Avoid

A.6. Meeting Notes

Participants	Hj. Wenger (Instructor), Frank Gauss, Lukas von Allmen
Agenda	- Goals presentation - Clarification whether there is a deadline or submission required for the presentation (besides adding it to the addendum) - Clarification whether the defense presentation should be an enhanced version of the public presentation or a separate, more technical presentation - Proposal to establish a recurring meeting series every Thursday at 3 pm during the semester
Tasks	- Deliver a set of success criteria - Risk assessment

Table A.1.: Summary Checkpoint 1

Meeting Notes Nsak Checkpoint - 26.02.26

Participants	Hj. Wenger (Instructor), Urs Keller (Expert), Frank Gauss, Lukas von Allmen
Agenda	- Review of the project planning. - Discussion on the investigation and analysis of network communication between end devices and external networks (RED scenario: sniffing, filtering, and potential data exfiltration).
Decisions	- Variant 1 was selected for further development. - Open questions remain regarding the use and management of AI tokens.
General Notes	- A misunderstanding how to display the success criteria of the project leads to a goal section in the thesis instead of the goals in the milestone descriptions. - Discussion on the rationale behind selecting Variant 1. Consideration of whether a user interface is required when MCP functionality is available - Open Research Question. - NSAK integrates AI capabilities. - The development of a Web GUI is currently considered lower priority - suggestion of Urs Keller.

Table A.2.: Summary Checkpoint2

Meeting Notes Nsak Checkpoint - 05.03.26

Meeting Notes Nsak Checkpoint - 19.03.26

Meeting Notes Nsak Checkpoint - 02.04.26

Meeting Notes Nsak Checkpoint - 16.04.26

Meeting Notes Nsak Checkpoint - 30.04.26

Meeting Notes Nsak Checkpoint - 14.05.26

Meeting Notes Nsak Checkpoint - 28.05.26

Meeting Notes Nsak Checkpoint - 11.06.26

A.7. Self-Hosted LLM

During the A.5.1 we realized that the costs for AI tokens required to use cloud based LLMs might cause huge costs. As the project team already has existing hardware which is suitable for running smaller LLMs, one of the possibilities to avoid this risk is the usage of self-hosted LLMs. Besides eliminating the costs for AI tokens, at least during the development phase, it also allows the quick evaluation of different AI Models. Depending on the quality and performance of the smaller models running in the self-hosted setup we can consider not using a cloud based LLM at all.

Add source for the cost claim

A.7.1. Hardware Setup

Component	**Specification**
CPU	AMD Ryzen 5 3600 (6 cores / 12 threads, up to 4.2 GHz)
GPU	NVIDIA GeForce RTX 3070 (8 GB VRAM)
System Memory	32 GB RAM

Table A.3.: Hardware configuration of the self-hosted LLM inference server.

The 8 GB VRAM of the RTX 3070 allows us to run models with 7 Billion parameters

A.7.2. System Setup

LLM Model Manager - Ollama

Webserver - Nginx

LLM Chat GUI - Open WebUI

System Hardening

```
# Install UFW
sudo pacman -Syu ufw
sudo ufw status

# Allow all outgoing connections
sudo ufw default allow outgoing

# Deny all incoming connections per default
sudo ufw default deny incoming

# Allow SSH for remote access
sudo ufw allow ssh

# Allow HTTP for certbot HTTP challange
sudo ufw allow http

# Allow HTTPS for Open WebUI
sudo ufw allow https

# Allow default Ollama port
sudo ufw allow 11434/tcp

# Make sure to run this command after adding the "allow" rules!
sudo ufw enable

sudo ufw status
```

Listing A.1: UFW configuration

Port Forwarding via Fritzbox DynDNS via Infomaniak

A.7.3. Evaluation